Case Study 9: Student Marks Matrix Analysis
Complete Python Source Code

```
students = ["Asha", "Ravi", "Noor"]
subjects = ["Python", "Maths", "English"]
marks = [
    [80, 75, 90],
    [65, 85, 70],
    [45, 55, 60]
]
low_average_limit = 50
print("STUDENT TOTALS AND AVERAGES")
totals = []
for i in range(len(students)):
    total = sum(marks[i])
    average = total / len(subjects)
    totals.append(total)
    print(f"{students[i]}: total={total}, average={average:.2f}")
print("\nSUBJECT AVERAGES")
subject_averages = []
for j in range(len(subjects)):
    column_total = 0
    for i in range(len(students)):
        column_total += marks[i][j]
    average = column_total / len(students)
    subject_averages.append(average)
    print(f"{subjects[j]}: {average:.2f}")
top_index = totals.index(max(totals))
print(f"Top performer: {students[top_index]} ({totals[top_index]} marks)")
low_subjects = []
for j in range(len(subjects)):
    if subject_averages[j] < low_average_limit:
        low_subjects.append(subjects[j])
print("Subjects below average threshold:", low_subjects if low_subjects else "None")
```